Homework 1

1. (20 points) **Perfect Secrecy** (Ch. 1–2)

   (a) (6 points) Consider a modified one-time pad where we use the bitwise AND ($\wedge$) operation instead of XOR ($\oplus$):

   $\mathsf{Enc}(K, M) = K \wedge M$ and $\mathsf{Dec}(K, C) =?$

   i. Is this scheme correct? If yes, specify the decryption function. If not, explain why.

   ii. Does this scheme provide perfect secrecy? Justify your answer.

   (b) (8 points) Consider the following variant of a one-time pad operating on decimal digits (0–9):

   $\mathsf{Enc}(K, M) = (K + M) \bmod 10$ and $\mathsf{Dec}(K, C) = (C - K) \bmod 10$

   where $K, M, C \in \{0, 1, 2, \ldots, 9\}$.

   i. Prove that this scheme is correct.

   ii. Prove that this scheme provides perfect secrecy, assuming K is chosen uniformly at random.

   (c) (6 points) Consider a one-time pad where the key length is half the message length: $\mathsf{Enc}(K, M) = (K \oplus M_1, K \oplus M_2)$ where $M = (M_1, M_2)$ and $|M_1| = |M_2| = |K|$.

   Provide a specific attack that breaks the confidentiality of this scheme, showing clearly the information an attacker can extract from the ciphertext.

2. (20 points) **Libraries and Indistinguishability** (Ch. 2)

   (a) (10 points) Consider the following libraries:

   $\mathcal{L}_1$

   $\textsc{init}()$:
   $K \twoheadleftarrow \{0,1\}^n$

   $\textsc{query}(M)$:
   return $K \oplus M$

   $\approx$

   $\mathcal{L}_2$

   $\textsc{init}()$:
   $R_1 \twoheadleftarrow \{0,1\}^n$
   $R_2 \twoheadleftarrow \{0,1\}^n$

   $\textsc{query}(M)$:
   if $M = R_1$ return R_2
   else return $M \oplus R_1 \oplus R_2$

   Are these libraries indistinguishable? Either prove they are indistinguishable or provide a distinguisher program that can tell them apart with non-negligible probability.

   (b) (10 points) State whether the following libraries are indistinguishable and justify your answer:

   $\mathcal{L}_\mathsf{C}$

   $K \twoheadleftarrow \{0,1\}^n$

   $\textsc{enc}(M)$:
   $C := K \oplus M$
   return C

   $\textsc{dec}(C)$:
   $M := K \oplus C$
   return M

   $\approx$

   $\mathcal{L}_\mathsf{D}$

   $\textsc{enc}(M)$:
   $C \twoheadleftarrow \{0,1\}^n$
   return C

   $\textsc{dec}(C)$:
   $M \twoheadleftarrow \{0,1\}^n$
   return M

3. (20 points) **Security Proofs** (Ch. 2)

Let $\Sigma = (\mathsf{KeyGen}, \mathsf{Enc}, \mathsf{Dec})$ be a secure one-time pad encryption scheme for messages in $\{0,1\}^n$. Consider the following modified scheme $\Sigma' = (\mathsf{KeyGen}', \mathsf{Enc}', \mathsf{Dec}')$:

$$\begin{aligned} \mathsf{KeyGen}'() &= K \twoheadleftarrow \mathsf{KeyGen}() \\ \mathsf{Enc}'(K, M) &= (C_1, C_2) \text{ where } C_1 \twoheadleftarrow \mathsf{Enc}(K, M) \text{ and } C_2 \twoheadleftarrow \mathsf{Enc}(K, M \oplus 1^n) \\ \mathsf{Dec}'(K, (C_1, C_2)) &= \mathsf{Dec}(K, C_1) \end{aligned}$$

Determine whether Σ' is a secure encryption scheme. If it is secure, provide a formal proof. If it is not secure, describe a concrete attack that breaks its confidentiality and explain why the attack works.

4. (20 points) **PRGs and PRFs** (Ch. 5–6)

   (a) (10 points) Analyze the security implications of the following PRG construction, where G is a secure PRG:

   $$H(S) = A \| B \| C \| D \text{ where } A \| B = G(S) \text{ and } C \| D = G(B)$$

   Determine whether H is a secure PRG. If not, provide a distinguisher that can tell $H(S)$ apart from a truly random string with non-negligible advantage.

   (b) (10 points) Consider the following PRF construction: $F(K, X) = G(K) \oplus X$, where G is a secure PRG. Is F a secure PRF? If not, describe a distinguisher that can tell F apart from a random function with non-negligible advantage.

5. (20 points) **CPA and CCA Security** (Ch. 8–9)

   (a) (10 points) For each of the following encryption schemes, determine whether it achieves CPA security. If not, provide a specific attack:

      i. $\mathsf{Enc}(K, M) = (R, F(K, R) \oplus M)$ where $R \twoheadleftarrow \{0,1\}^\lambda$ and F is a secure PRF.
      ii. $\mathsf{Enc}(K, M) = (R, F(K, M) \oplus R)$ where $R \twoheadleftarrow \{0,1\}^\lambda$ and F is a secure PRF.
      iii. AES in Electronic Codebook (ECB) mode.
      iv. AES in Counter (CTR) mode with a randomly chosen IV.

   (b) (10 points) For the following encryption scheme constructions, determine whether each provides CCA security and/or authenticated encryption (AE). Justify your answers with brief explanations:

      i. Encrypt-then-MAC: $C = \mathsf{Enc}(K_e, M)$, $T = \mathsf{MAC}(K_m, C)$, output (C, T)
      ii. Encrypt-and-MAC: $C = \mathsf{Enc}(K_e, M)$, $T = \mathsf{MAC}(K_m, M)$, output (C, T)
      iii. MAC-then-encrypt: $T = \mathsf{MAC}(K_m, M)$, $C = \mathsf{Enc}(K_e, M \| T)$, output C